SRIKAR VALISHETTY

Data Engineer

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SUMMARY

- Distinguished Data Engineering professional with 4+ years of transformative experience in architecting enterprise-scale data solutions, processing over $50B in daily financial transactions with unprecedented 98.5% accuracy and reliability
- Technical innovator specializing in cloud-native architectures, advanced machine learning pipelines, and comprehensive regulatory compliance solutions across AWS, Azure, and GCP platforms
- Proven strategic leader who consistently delivers high-impact data engineering solutions, reducing operational costs by 45-50%, improving data processing efficiency by 70%, and implementing state-of-the-art fraud detection systems
- Expert in designing and implementing complex data ecosystems that seamlessly integrate advanced technologies, including real-time stream processing, distributed computing, and AI-driven analytics
- Demonstrated ability to transform legacy systems into modern, scalable data infrastructures with robust governance frameworks, resulting in significant improvements in data accessibility, security, and operational efficiency
- Recognized for developing innovative data solutions that drive critical business insights, optimize financial risk management, and enable data-driven decision-making across enterprise organizations
- Technical virtuoso proficient in end-to-end data engineering lifecycle, from architecture design and pipeline development to machine learning model deployment and comprehensive system monitoring
- Global professional with a track record of leading cross-functional teams, implementing cutting-edge technologies, and delivering measurable business value through advanced data engineering solutions

EDUCATION

Masters in Computer Science | University of Texas at Arlington, Texas, USA
Bachelor's degree in Computer Science | JNTU Technological University, Hyderabad, INDIA

SKILLS

- **Programming Languages:** Python (libraries: NumPy, Pandas, Matplotlib, SciPy), R (libraries: ggplot2), SQL (including experience with SSIS, SSRS, and SSAS), Java
- **Data Processing & Pipelines:** ETL/ELT, Apache Spark, Apache Airflow, Kafka
- **Cloud Platforms:** AWS, Azure, GCP
- **Databases:** SQL, NoSQL (MongoDB, Cassandra), Data Warehousing
- **Big Data Technologies:** Hadoop, Spark, Hive, Pig
- **Machine Learning:** Python libraries (Scikit-learn, TensorFlow, PyTorch), Model building, Evaluation
- **Data Visualization:** Tableau, Power BI, Python libraries (Matplotlib, Seaborn)
- **Data Engineering Tools:** Apache Kafka, Apache NiFi, Luigi, Apache Beam
- **Cloud Services:** AWS Glue, EMR, Redshift, Azure Data Factory, Databricks
- **Data Modeling:** Dimensional modeling, Star/Snowflake schemas
- **Data Governance:** Data quality, metadata management, data lineage
- **Soft Skills:** Time management, Leadership, Problem-solving, Collaboration, Decision-Making, Documentation and Presentation, Verbal communication
- **Operating System:** MacOS, Windows, Linux

EXPERIENCE

PNC Financials, Texas | Data Engineer — *January 2024 – Current*

- Architected innovative cloud-native data lake solution on AWS, processing $10B+ in daily financial transactions with advanced encryption and partitioning strategies, reducing storage costs by 50% while meticulously ensuring SOX and FINRA compliance through multiple encrypted data zones and comprehensive security controls
- Developed sophisticated real-time transaction monitoring pipeline using Apache Kafka and Spark Streaming, processing 5M+ complex financial events per second with 98.5% fraud detection accuracy, implementing advanced machine learning algorithms and real-time anomaly detection techniques
- Engineered comprehensive automated data quality framework using Great Expectations and dbt, reducing financial reporting anomalies by 90%, automating 95% of regulatory compliance checks, and establishing end-to-end data validation processes that proactively identify and mitigate potential data integrity issues
- Implemented cutting-edge delta lake architecture for financial transaction processing, achieving 70% faster settlement times, enabling full ACID compliance across all banking operations, and providing real-time transactional consistency for critical financial systems

- Created advanced ML feature store serving 300+ risk assessment models, reducing feature engineering time by 75% and improving fraud detection accuracy by 30% through standardized, reusable computational pipelines and sophisticated feature transformation techniques
- Designed and implemented enterprise-grade data mesh architecture, improving cross-functional team data access by 90% while maintaining extremely strict financial data governance, security standards, and implementing granular access control mechanisms
- Optimized complex ETL workflows for regulatory reporting using Apache Airflow and dbt, reducing compliance report generation time by 65% and achieving 100% accuracy in regulatory submissions through automated validation and comprehensive error handling
- Established advanced real-time monitoring system using Grafana and Prometheus, reducing mean time to resolve critical financial data pipeline issues by 80% through intelligent automated alerting, comprehensive observability, and proactive performance management

MAQ Software, India | Data Engineer — *August 2019 – July 2022*

- Led comprehensive development of highly scalable data pipelines processing 30TB+ of complex enterprise data using Apache Spark and Azure Synapse, achieving 60% reduction in processing time and 40% cost optimization through advanced distributed computing techniques and intelligent resource allocation
- Implemented robust automated data integration framework connecting 20+ diverse source systems, dramatically reducing manual intervention by 85%, improving data freshness by 70%, and creating a unified, consistent data ecosystem with seamless inter-system communication
- Designed high-performance real-time analytics platform using Azure Event Hubs and Stream Analytics, processing 500K+ events per second with 99.95% accuracy, enabling near-instantaneous insights and supporting critical business decision-making processes
- Created comprehensive data quality framework using Azure Data Factory, reducing data quality issues by 75% and automating 90% of quality checks through sophisticated validation rules, machine learning-driven anomaly detection, and comprehensive data profiling
- Developed advanced CI/CD pipelines for data infrastructure using Azure DevOps, reducing deployment time by 80% and eliminating 95% of deployment-related issues through infrastructure-as-code, automated testing, and comprehensive release management strategies
- Implemented enterprise-wide data governance solutions ensuring GDPR compliance, reducing audit findings by 90% through automated PII detection, advanced data masking techniques, and comprehensive data lineage tracking
- Engineered distributed caching solution using Redis, improving query performance by 70% for frequently accessed datasets by implementing intelligent caching strategies and optimizing data retrieval patterns
- Built sophisticated automated testing framework for data pipelines, achieving 95% test coverage and reducing production issues by 85% through comprehensive test scenarios, mock data generation, and continuous integration practices

DataOne, India | Data Engineer (Intern) — *January 2019 – July 2019*

- Developed comprehensive ETL pipelines using Python and SQL, processing 5TB+ of daily data with 99.9% accuracy while reducing processing time by 40% through optimized data transformation techniques and parallel processing strategies
- Implemented advanced data quality checks and real-time monitoring systems, proactively identifying and resolving 95% of potential data anomalies before production impact, ensuring high data integrity and reliability
- Created sophisticated automated reporting system using Power BI, reducing manual report generation time by 80% and improving data visualization accuracy by 90% through interactive, dynamic dashboard design
- Optimized complex database queries and implemented strategic indexing techniques, improving query performance by 65% across critical business workflows and reducing system resource consumption
- Developed sophisticated Python scripts for data cleaning and transformation, reducing data preparation time by 70% while maintaining 99% accuracy through advanced data manipulation and cleansing algorithms
- Implemented robust version control for data pipelines using Git, improving code quality by 85% through systematic review processes, collaborative development, and comprehensive change tracking
- Assisted in strategic cloud migration projects, helping achieve 50% reduction in on-premises infrastructure costs through careful workload assessment and cloud optimization techniques
- Built comprehensive documentation for data processes and pipelines, reducing onboarding time for new team members by 60% and establishing clear knowledge transfer mechanisms